The authors developed a new four-dimensional topside model based on GNSS radio occultation data using L2-ANN method. This work is interesting and gives new thoughts on building better topside ionospheric model. The model is validated by out-of-sample COSMIC data, GRACE and ISR data, and compared to IRI-2016. Although the model seems reasonable according to the validation results, some issues should be discussed more or revised appropriately before potential publication in the journal.

1. Among the listed input variables, the year and day are not considered (or involved in the "decimal month"?), which is unusual when dealing with the ionospheric parameters by ANN method.
2. In line 192~199, the VSH is used as data pre-processing criterion. The scale height is one of the most significant parameters in the topside ionosphere, and many mathematical descriptions about the topside adopt the scale height as anchor point in the shape function. I wonder if the scale height is added as another input variable, the topside model is further developed or not.
3. In line 120~124, "where the hmF2 and NmF2 measurements is unavailable" (BTW, "are unavailable"), the sub-models are used for providing the inputs. But for RO measurements, how often dose this situation occur? According to table 5, the topside model is negatively influenced by adopting the sub-models (from 9.3% to 27.7%). More explanations about the necessity of using sub-models should be added here.
4. In the abstract, authors mentioned, "In comparison with these variables, F10.7 and Kp, together with hmF2 and NmF2, affect the performance in a smaller scale." In the text (table 3 and 4), the nine variables are added one by one into the modeling process, and the RMSE and relative residuals for each scheme are listed in table 4. There is no doubt that using more related input variables, the residuals keep decreasing. However, the order of adding variables shown in table 3 is just one of many possible ways. The significance of each variable cannot be revealed by the decrease amount of RMS or residuals. In my view, it is unreasonable to consider the latter variables less significant, since they are added to the system later.
5. In Figure 5, why are the RMSE for three datasets scattered a lot in some panels? In Figure 4, what are the small bumps in the residual plot of the altitude? The residual plots of latitude and hmF2 have dramatic increase at high latitudes and altitudes. Maybe those RO profiles should be discarded because of possible ionospheric disturbances and inaccurate hmF2 value.

Minor revisions:

1. Line 114: add explanation about L2-ANN when first mentioned;

2. Line 270: The RO-based hmF2 model is already available as alternative choice in current IRI model.
3. Line 342: “The altitude and time resolution is 50 km and 6 hours respectively.” The time resolution seems to be 2 hours according to latter figure captions.
4. Figure 6 is hardly recognized.
5. In table 3, the variable DoY is used while it is presented as ”month” in table 1. The variable used should be made consistent.